



Data Glacier

Your Deep Learning Partner

Exploratory Data Analysis

Project: Hate Speech Detection using Transformers (Deep Learning)
Data Glacier Virtual Internship (Week 11 Deliverable)

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Agenda

- Problem Description
- Dataset Information
- Exploratory Data Analysis(EDA)
- Final Recommendation

Problem Description

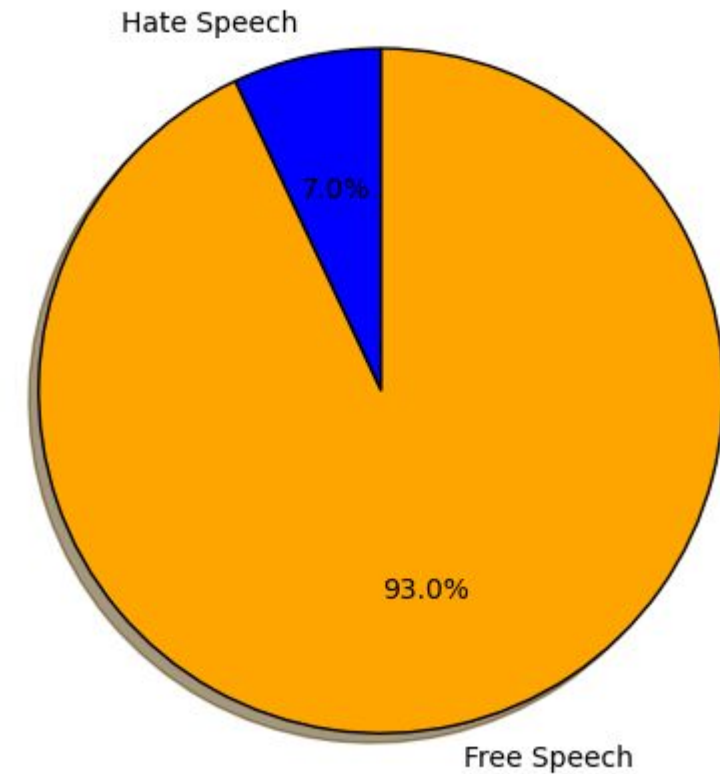
- Hate speech refers to any communication that attacks or discriminates against individuals or groups based on identity factors like religion, ethnicity, gender, or nationality. In this project, we aim to build a hate speech detection model using Machine Learning and Deep Learning techniques. We use Twitter data to classify whether a tweet contains hate speech or not.

Dataset Information

train_E6oV3lV.csv

This file contains a list of tweet, labeled as hate or free speech.

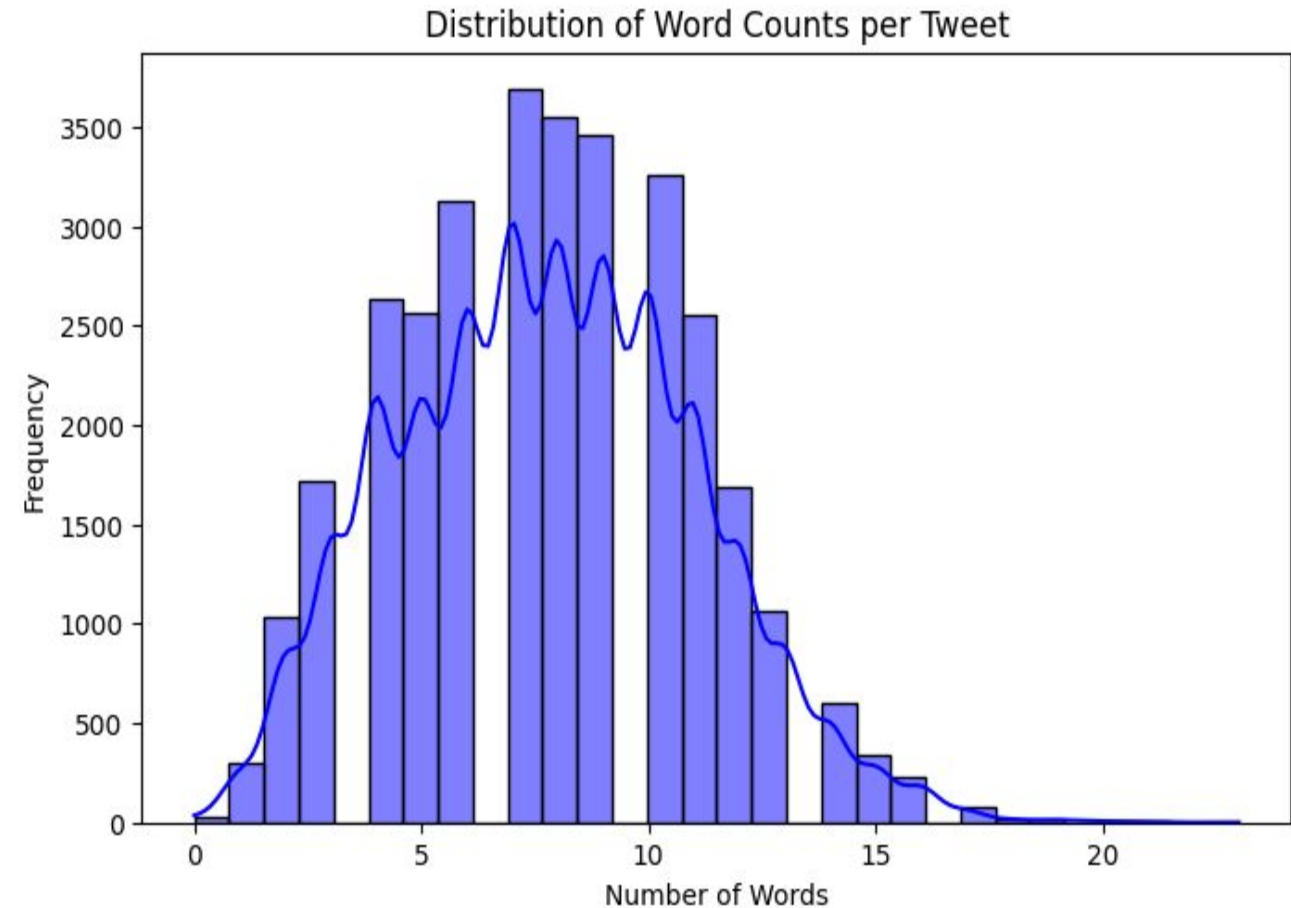
Distribution of Hate vs Free Speech



EDA

Number of Words per Tweet

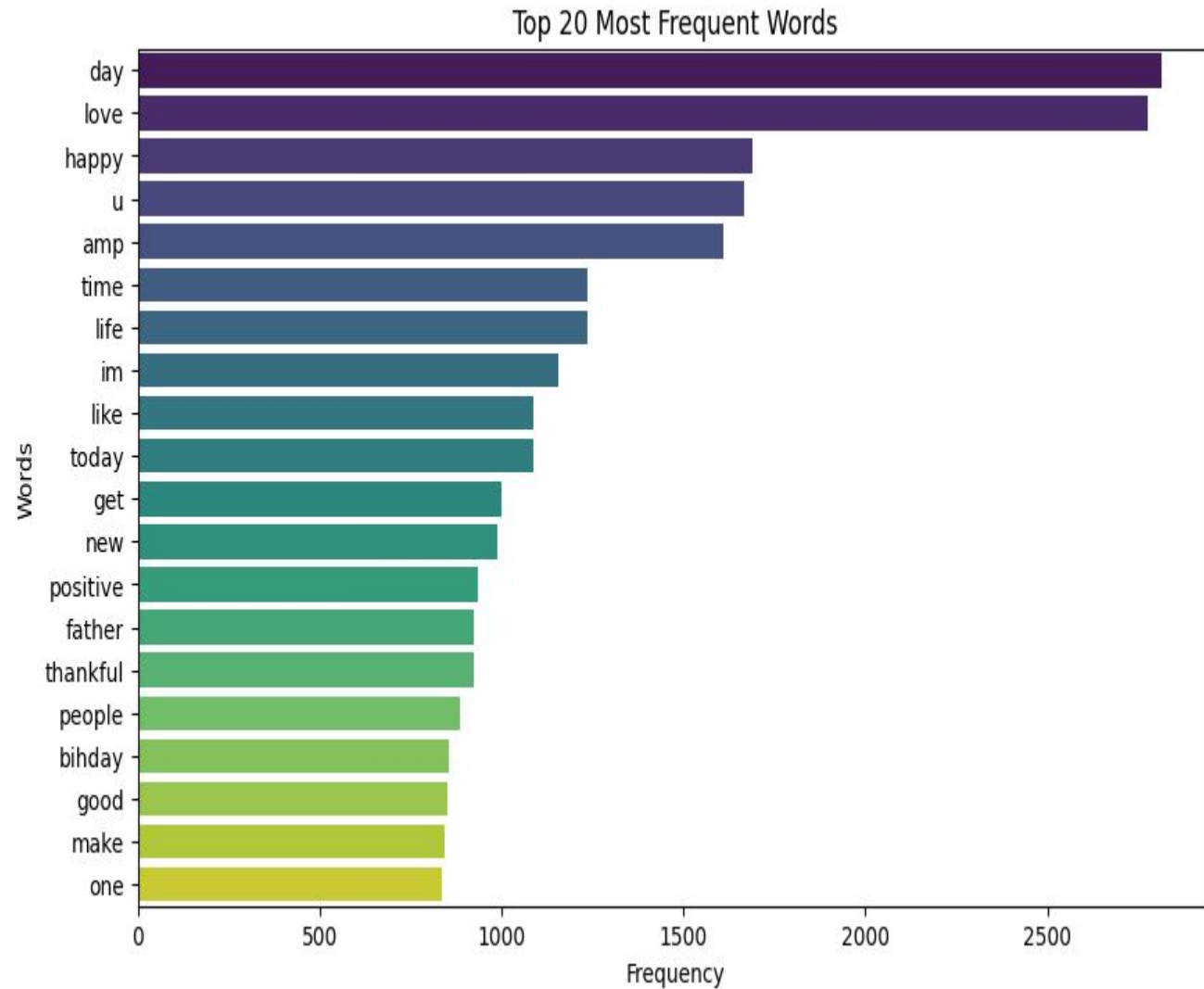
- The majority of tweets contain between 5 to 15 words.



EDA

Most Frequent Words in the Whole Dataset

- The 20 most frequently occurring words across the entire dataset.

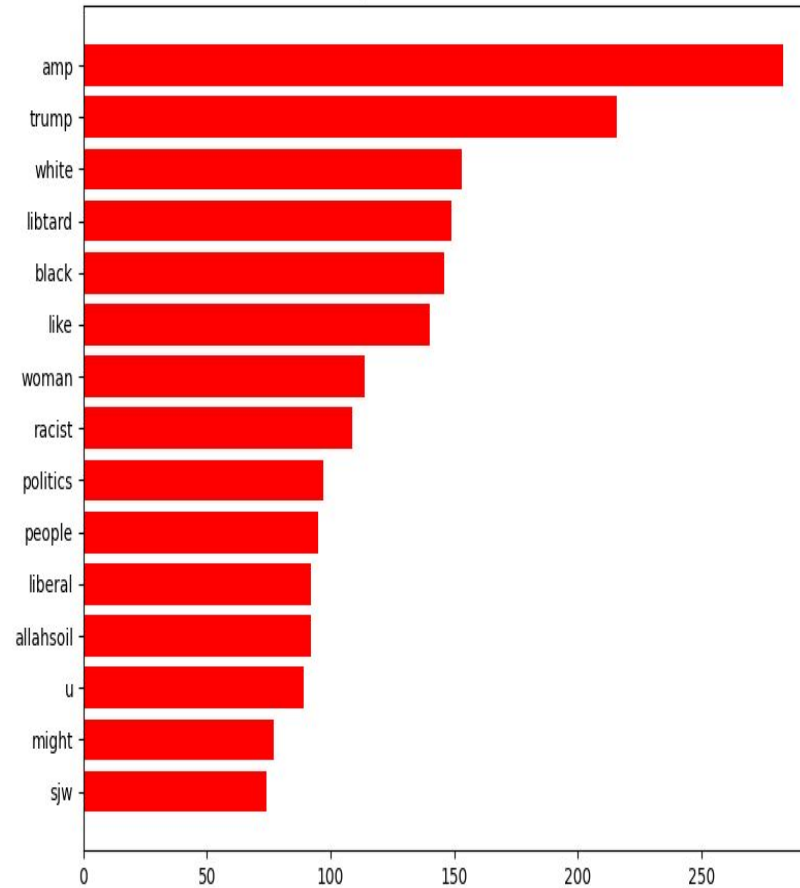


EDA

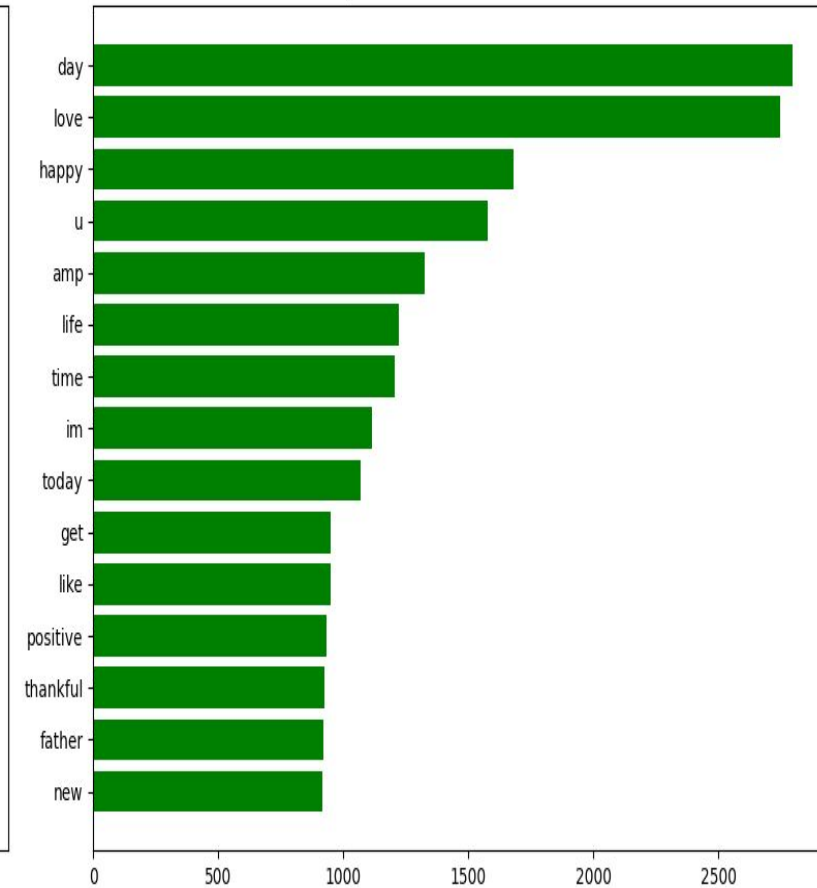
Most Frequent Words in Hate vs. Non-Hate Tweets

- Hate tweets tend to use more aggressive or targeted language, while non-hate tweets exhibit more neutral or conversational vocabulary.

Top Words in Hate Tweets



Top Words in Non-Hate Tweets



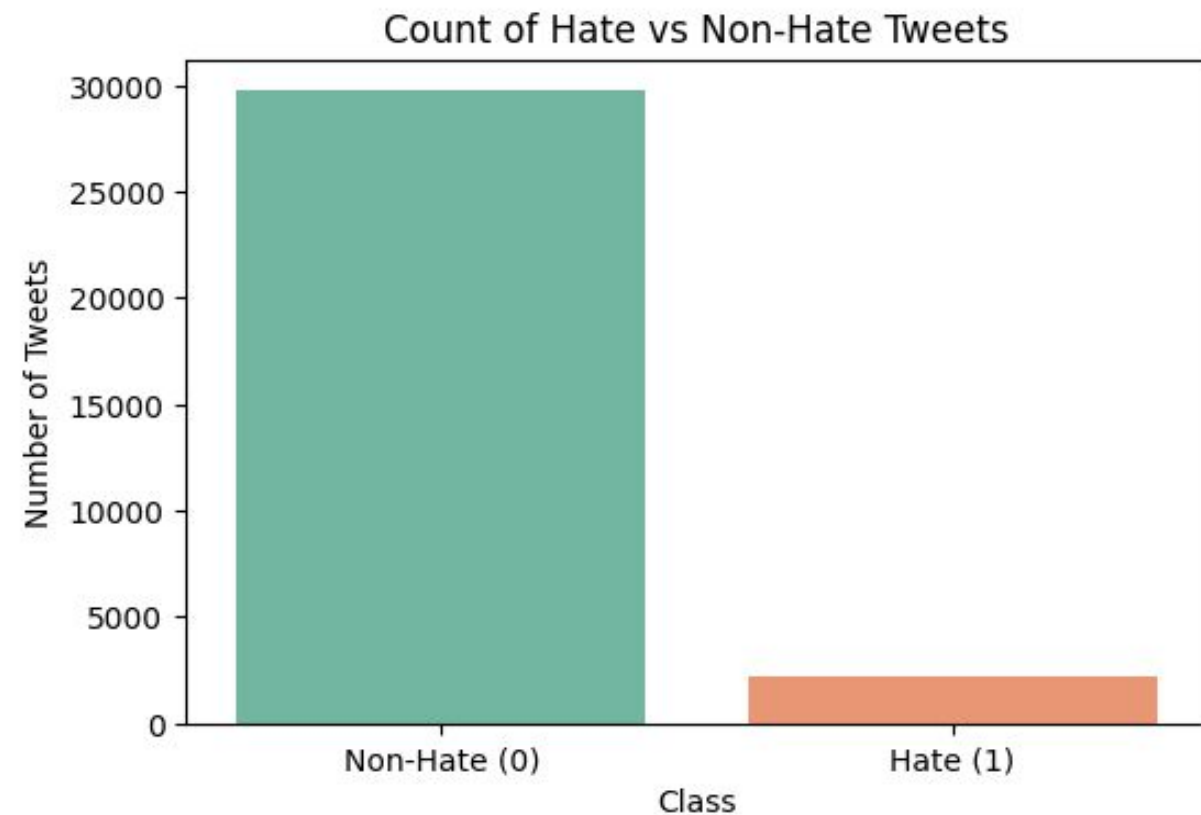
Word Cloud for Each Level

- [illegible]

EDA

Count of Hate Speech vs. Non-Hate Speech

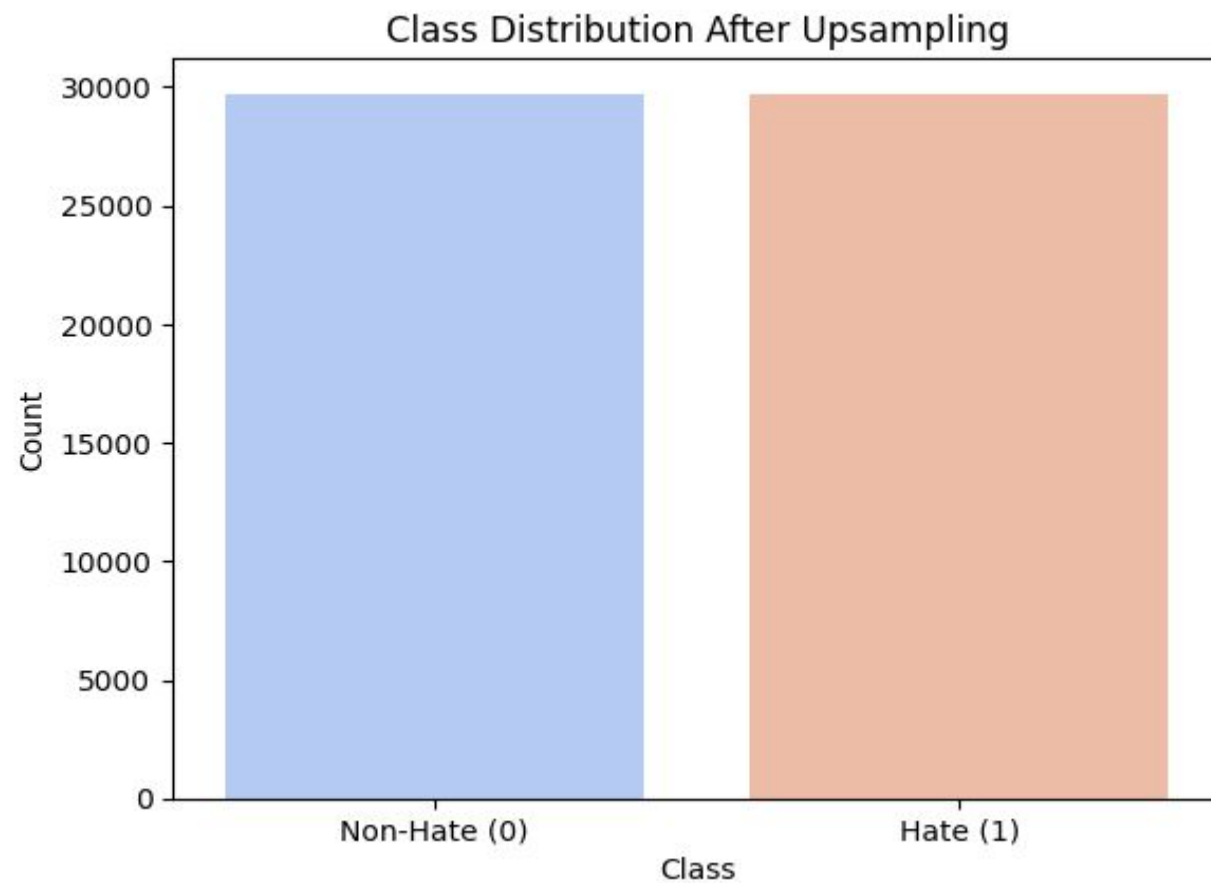
- The original class distribution is visualized.



EDA

Solve the Imbalance Issue

- Applied upsampling to the minority class (hate tweets).





Recommended Models & Feature Strategies

Model	Type	Feature Extraction Method	Reason
Random Forest	Traditional ML	TF-IDF	Handles high-dimensional sparse vectors well
XGBoost	Traditional ML	TF-IDF	Robust to overfitting, great for classification
CNN+LSTM	Deep Learning	Word Embeddings	CNN extracts spatial features, LSTM models sequence
Transformer (BERT)	Deep Learning	BERT Embeddings	State-of-the-art contextual modeling

Thank You